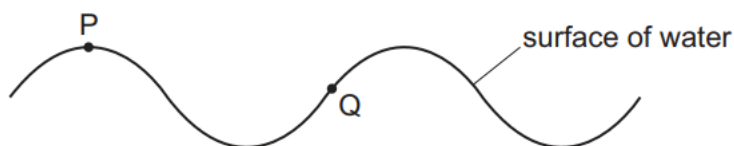
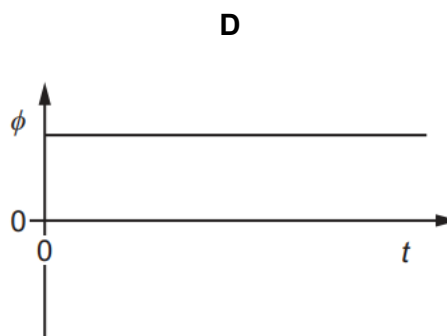
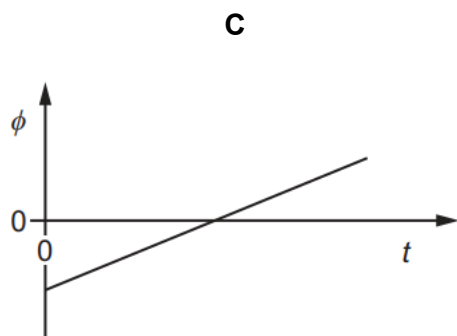
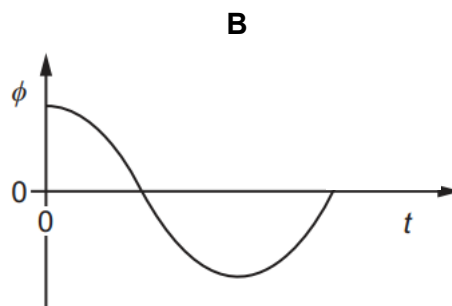
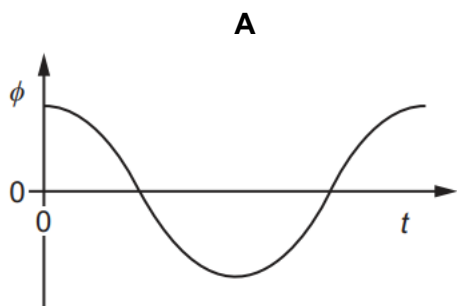


- 18 In a progressive water wave, two particles P and Q, on the surface of the water, are a fixed horizontal distance apart. P and Q oscillate vertically.

At time $t = 0$, the wave is as shown.



Which graph best represents the variation with time t of the phase difference ϕ between the oscillation of the water particle P and the oscillation of the water particle Q?



Ans: D

$$\Delta\phi = \frac{\Delta x}{\lambda} 2\pi \text{ is independent of } t.$$

